INSTITUTO SUPERIOR TECNOLÓGICO QUITO

Excelencia en Educación Superior

5.1 Clasificacion

python™

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In [1]:
```
import numpy as np
from math import sqrt
from pprint import pprint
from sklearn import datasets, linear_model, metrics
from sklearn.dummy import DummyRegressor
from sklearn.model_selection import cross_validate, KFold, cross_val_predict, train
from sklearn import preprocessing
from sklearn.metrics import make_scorer, mean_squared_error
import matplotlib.pyplot as plt
```

In [2]:
```
# Carga de datos.
#datos = datasets.load_boston()
datos = datasets.fetch_california_housing()
#print(datos)
X = datos.data
y = datos.target
print('Dimensiones de X: ', np.shape(X))
# print(y)
```

```
Dimensiones de X:  (20640, 8)
```

In [3]:
```
# # Métricas de evaluación.
metricas = {
  'MAE':  'neg_mean_absolute_error',
  'RMSE': make_scorer(lambda y, y_pred:
                      sqrt(metrics.mean_squared_error(y, y_pred)),
```

```
                        greater_is_better=False),
  'MAPE': make_scorer(lambda y, y_pred:
                        np.mean(np.abs((y - y_pred) / y)) * 100,
                        greater_is_better=False),
  'R2':   'r2',}
```

In [4]:
```
# 1) Partición de datos externa
X_training, X_testing, y_training, y_testing = train_test_split(X, y, test_size=0.2
print([np.shape(X_training), np.shape(X_testing)])
```

```
[(16512, 8), (4128, 8)]
```

In [5]:
```
# 2) Extracción de características
# 3) Selección de atributos
# 4) Estandarización de los datos de entrenamiento
standardizer = preprocessing.StandardScaler()
stdr_trained = standardizer.fit(X_training)
X_stdr = stdr_trained.transform(X_training)
```

In [6]:
```
# 5) Construcción del algoritmo de aprendizaje.
reg = linear_model.LinearRegression(fit_intercept=True)
```

In [7]:
```
# 5.1) Validación cruzada interna

# Extraer directamente resultados del error para cada bolsa, en lugar de las predic
cross_val_results = cross_validate(reg, X_stdr, y_training,
                                   cv = KFold(n_splits=5, shuffle=True, random_sta
#print("cross_val_MAE:   %0.4f +/- %0.4f" % (-cross_val_results.mean(), cross_val_r
pprint(cross_val_results)
```

```
{'fit_time': array([0.06502652, 0.00558782, 0.00792241, 0.00758338, 0.00607681]),
 'score_time': array([0.00404501, 0.00357199, 0.00200224, 0.00458193, 0.00262785]),
 'test_MAE': array([-0.54071407, -0.52878981, -0.51274193, -0.53512422, -0.5279336
4]),
 'test_MAPE': array([-31.74769388, -31.65516613, -30.97054077, -32.61432353,
       -30.685065  ]),
 'test_R2': array([0.60970239, 0.60411343, 0.6354319 , 0.60076148, 0.60727452]),
 'test_RMSE': array([-0.73389779, -0.72516701, -0.69726867, -0.73337185, -0.7128422
5])}
```

In [8]:
```
# 5.2) Entrenamiento con todos los datos para obtener los coeficientes del modelo.
model = reg.fit(X_stdr, y_training)

# Obtención de los coeficientes del modelo.
w = model.coef_
print('Model coeficients: \n', w)

# Obtención del término independiente del modelo.
w_0 = model.intercept_
print('\nTérmino independiente: ', w_0)
```

```
Model coeficients:
 [ 0.85438303  0.12254624 -0.29441013  0.33925949 -0.00230772 -0.0408291
 -0.89692888 -0.86984178]

Término independiente:  2.071946937378619
```

In [9]:
```
# ---- PREDICCIÓN ---- #
```

In [10]:
```
# 6) Extracción de las características de test
# 7) Selección de los atributos de test
# 8) Estandarización de las característiacs de test
X_test_stdr = stdr_trained.transform(X_testing)
```

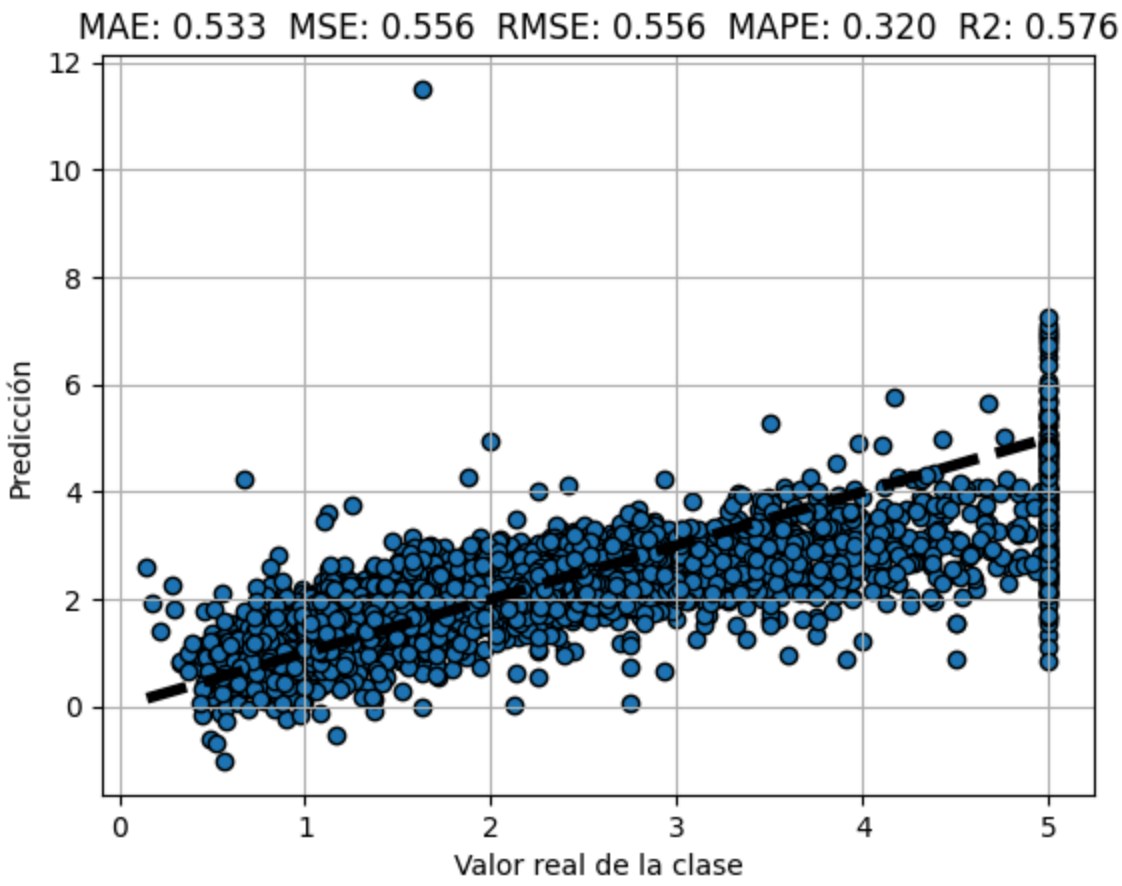
In [11]:
```
# 9) Predicción del conjunto de test
y_pred_test = model.predict(X_test_stdr)
```

In [12]:
```
# Cálculo de las métricas de evaluación.
MAE = metrics.mean_absolute_error(y_testing, y_pred_test)
MSE = metrics.mean_squared_error(y_testing, y_pred_test)
RMSE = metrics.mean_squared_error(y_testing, y_pred_test)
MAPE = metrics.mean_absolute_percentage_error(y_testing, y_pred_test)
R2 = metrics.r2_score(y_testing, y_pred_test)

print('MAE:  %.4f' % MAE)
print('MSE: %.4f' % MSE)
print('RMSE: %.4f' % RMSE)
print('MAPE: %.4f' % MAPE)
print('R2:   %.4f' % R2)
```

```
MAE:  0.5332
MSE: 0.5559
RMSE: 0.5559
MAPE: 0.3195
R2:   0.5758
```

In [13]:
```
# Gráfica de realidad vs. predicción.
fig, ax = plt.subplots()
ax.scatter(y_testing, y_pred_test, edgecolors=(0, 0, 0))
ax.plot([y.min(), y.max()], [y.min(), y.max()], 'k--', lw=4)
ax.set_xlabel('Valor real de la clase')
ax.set_ylabel('Predicción')
plt.title("MAE: %.3f  MSE: %.3f  RMSE: %.3f  MAPE: %.3f  R2: %.3f" %
          (MAE,MSE,RMSE,MAPE,R2))
plt.grid()
plt.show()
```



GitHub

https://github.com/Erick-305/machine-learning.git